Week – 2:

I. Dataset Identification & Collection Strategy

(To be completed after studying Section IV of the research paper)

- Primary Dataset Name & Source:
- Dataset Specifications (Size, Format, License, Download URL):
  - *Format & Size:*
  - *License & Source URL:*
- Class Distribution & Duration Statistics (According to paper):
- Author-Specified Train / Validation / Test Splits:
- Alternative or Additional Datasets Considered:

II. Dataset Organization & Preprocessing Specifications

(To be completed during data pipeline design)

- Folder Architecture Established:
  - data/raw/, data/processed/, data/splits/ initialized? [ ] Yes / [ ] No
- Metadata Schema (CSV Tracking):
  - *Attributes tracked (filename, label, duration, split, source):*
- Class Imbalance Mitigation Strategy:
  - *Approach selected (oversampling / undersampling / re-weighting):*
- Paper's Preprocessing Pipeline Breakdown:
  - *Resampling rate:*
  - *Window size & Hop length:*
  - *Normalization method & Feature type (mel / CQT / MFCC):*

III. Practical Implementation & Verification Checklist

(Hands-on execution log for Week 2 based on your workflow plan)

- [ ] Downloading & Verifying Datasets: Downloaded primary dataset; performed integrity checks (file count, total duration, corrupt file filter); generated statistical histograms (clip durations, class balance, SNR distribution) and saved dataset_stats.py report.
- [ ] Selecting & Finalizing Dataset Splits: Implemented exact train/val/test splits matching the paper with fixed random seeds; verified zero data leakage (ensuring speaker/artist overlap is prevented); computed split statistics and saved manifest CSV files.
- [ ] Preprocessing Pipeline Implementation: Coded the full preprocessing pipeline (audio → feature extraction → normalization → saving as .npy or .pt); tested with 10 sample files; verified shape alignment with model input specifications and profiled processing speed (samples/sec).
- [ ] Augmentation Implementation & DataLoader Optimization: Implemented paper's data augmentations (pitch shift, time stretch, noise addition, SpecAugment frequency/time masking); benchmarked augmented vs. original spectrograms side-by-side; finalized PyTorch DataLoader with augmentations ensuring throughput ≥ 10 batches/sec.

IV. Implementation Notes & Code Snippets

- Dataset Verification Observations / Anomalies Found:
- Core Preprocessing / Augmentation Code Snippet: